

EXECUTIVE TECHNICAL REPORT & PROJECT SUMMARY

Campus Emergency & Smart Patient Triage System

A unified, mission-critical incident dispatch command platform merging rapid campus tactical reporting with automated heuristic triage and dynamic medical facility routing.

Status:	Local URL:	Architecture:	Persistence:
Production-Ready v2.1	http://localhost:5173	Role-Gated Hybrid	LocalStorage Engine

1. EXECUTIVE SUMMARY & PROBLEM SCOPE

The **Campus Emergency & Smart Patient Triage System** provides an enterprise-grade emergency response command platform designed specifically for university and institutional environments. The system bridges the critical time gap between incident occurrence and professional intervention by unifying two vital domains:

- **Tactical SOS Reporting (Caller Domain):** One-tap panic triggers, rich symptom reporting, and zero-delay bystander guidance.
- **Smart Patient Triage Engine (Clinical Domain):** Automated keyword heuristics, risk-factor age escalation, unit dispatch assignments, and localized facility routing.
- **Incident Telemetry Command Center (Admin Domain):** Real-time situational awareness, audio siren alerts, priority filtering, and ticket status progression (Dispatched → En Route → Resolved).

2. SYSTEM MODULES & FUNCTIONAL SCOPE

Dual-Role Authentication System

Secure role-gated access with distinct views for **Students/Callers** (locked to Tactical SOS Portal & facility guides) and **Administrators** (full access to Command Center, ticket controls, and telemetry).

1-Tap Zero-Delay Emergency Reporting

Rapid 1-tap preset triggers for **Cardiac/Collapse**, **Severe Trauma/Bleeding**, **Active Fire/Smoke**, and **Physical Threat/Assault** with instant dispatch and no manual typing required.

Heuristic Smart Triage Engine

Rule-based semantic parser categorizing incidents into **EMERGENCY**, **URGENT**, and **NORMAL**. Generates medical justifications and bystander first-aid checklists.

Campus Nearest Facilities Locator

Proximity engine mapping 25 campus locations to the closest **Medical Dispensary**, **AED Defibrillator Hub**, and **First-Aid Kiosks/Eyewash Stations** with real-time walking distances and ETAs.

3. RULE-BASED TRIAGE CLASSIFICATION LOGIC

PRIORITY	KEY SYMPTOM INDICATORS	UNITS DISPATCHED	IMMEDIATE PROTOCOL
EMERGENCY	Unconscious, cardiac arrest, chest pain, not breathing, severe bleeding, skull trauma, fire/smoke, anaphylaxis	Campus Medical Unit (CMU-1), Campus Ambulance, Security Escort	Audio siren alert ping, instant bystander CPR/pressure guide, dispatch in < 3 min
URGENT	Bone fracture, deep cut/laceration, high fever (> 103°F), fainting, asthma attack, eye chemical burn, age ≤5 or ≥65	First-Aid Mobile Responder, Campus Medical Unit (CMU-2)	Immobilization guide, continuous vitals monitoring, prompt medical evaluation
NORMAL	Mild headache, minor abrasion, sore throat, general malaise, routine inquiry	Campus Dispensary Staff, Facilities Desk	Routing to Health Center Block C Room 102 for walk-in assessment

4. INTEGRATED CAMPUS EMERGENCY FACILITIES DIRECTORY

The system embeds a comprehensive campus infrastructure directory providing dynamic distance calculations from any selected building:

FACILITY HUB	BUILDING / LOCATION	KEY EQUIPMENT	CONTACT & STAFF
Central Medical Dispensary	Health & Wellness Block C, Room 102	Oxygen Cylinders, IV Fluids, Suture Kits, Trauma Stretcher, Cardiac Monitor	Dr. Vikram Singh (Duty CMO) • Ext. 402
Library AED Hub 1	Central Library Main Foyer	AED Defibrillator Model CR2, CPR Pads, Trauma Scissors	Ext. 108 (24/7 Security Foyer)
Sports Complex AED Hub 2	Sports Complex Physiotherapy Bay	AED Defibrillator Zoll 3, Spine Board, Ice Splints	Ext. 305 (Physio Coordinator)
Block A First-Aid & Eyewash	Block A (Engineering) 2nd Floor	Chemical Eyewash Bottle, Burn Gel, Tourniquets	Ext. 112 (Block A Desk)
Block B Trauma Kiosk	Block B 1st Floor East Lobby	Hemostatic Gauze, Cold Packs, Splint Kits	Ext. 210 (Block B Supervisor)
Workshop Safety & Deluge Shower	Mechanical Workshop Safety Bay 1	Emergency Deluge Shower, Fire Blankets, Burn Shield	Ext. 555 (Workshop Safety Desk)

5. PERSISTENT STORAGE & TICKET SYNCHRONIZATION MODEL

All incident records are maintained in browser-backed persistent local storage (campus_triage_incidents_v2) with multi-tab storage event synchronization.

- **Instant Write-Through:** Submissions and status transitions write to storage before UI update.
- **Reload Synchronization Actions:** Users receive explicit "*Reload Page to Update Ticket*" controls and an optional 3-second auto-reload countdown to synchronize live database state.
- **Command Center Telemetry Reload:** An administrative "*Reload Telemetry*" action in the dashboard header allows one-click data synchronization from persistent disk.

6. DEMONSTRATION ACCOUNTS & ACCESS CREDENTIALS

ROLE	USER NAME	INSTITUTIONAL EMAIL	PASSWORD	ASSIGNED SCOPE
STUDENT	Priya Menon	priya.menon@campus.edu	Student@123	SOS Emergency Portal, 1-Tap Trigger, Facility Locator
STUDENT	Rahul Sharma	rahul.sharma@campus.edu	Rahul@456	SOS Emergency Portal, 1-Tap Trigger, Facility Locator
ADMIN	Dr. Vikram Singh (CMO)	admin@campus.edu	Admin@2024	Full Command Center, Telemetry Feed, Ticket Operator

ROLE	USER NAME	INSTITUTIONAL EMAIL	PASSWORD	ASSIGNED SCOPE
ADMIN	Inspector Meera Krishnan	security@campus.edu	Security@999	Full Command Center, Telemetry Feed, Ticket Operator

7. TECHNOLOGY ARCHITECTURE & DIRECTORY LAYOUT

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campus-triage/ └─ src/ │ └─ lib/ │ │ └─ auth.js # Dual-role credential database & session manager │ │ └─ triage.js # Heuristic keyword engine & first-aid logic │ │ └─ facilities.js # Campus medical facilities & AED proximity directory │ │ └─ store.js # LocalStorage persistence & sync layer │ │ └─ mockData.js # 10 pre-loaded realistic incident cases │ └─ contexts/ │ │ └─ AuthContext.jsx # Role-based auth provider & hooks │ └─ components/ │ │ └─ Auth/LoginPage.jsx # Role selection, credential form & autofill │ │ └─ Layout/TopBar.jsx # Live clock, user badge & session actions │ │ └─ SOS/ # OneTapEmergency, NearestFacilitiesCard, SOSPortal, ResultCard │ │ └─ Dashboard/ # CommandCenter, MetricsBar, IncidentTable, IncidentRow, AlertRadar │ └─ hooks/useIncidents.js # Incident telemetry, Web Audio alerts & dispatch store │ └─ App.jsx # Role-gated route container
```